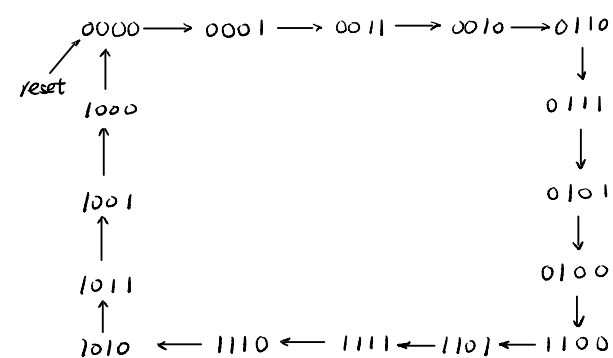


1. 解: ① 状态图



② 状态转换表

C_3	C_2	C_1	C_0	N_3	N_2	N_1	N_0
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	1
0	0	1	1	0	0	1	0
0	0	1	0	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	0	1	0	1
0	1	0	1	0	1	0	0
0	1	0	0	1	1	0	0
1	1	0	0	1	1	0	1
1	1	0	1	1	1	1	1
1	1	1	1	1	1	1	0
1	1	1	0	1	0	1	0
1	0	1	0	1	0	1	1
1	0	1	1	1	0	0	1
1	0	0	1	1	0	0	0
1	0	0	0	0	0	0	0

③ 卡诺图

1) N_3 :

$C_1 C_0$	00	01	11	10
$C_3 C_2$	00	0	0	0
01	1	0	0	0
11	1	1	1	1
10	0	1	1	1

$$N_3 = (C_2 + C_1 + C_0) \cdot (C_3 + C_0') \cdot (C_3 + C_1')$$

2) N_2 :

$C_1 C_0$	00	01	11	10
$C_3 C_2$	00	0	0	1
01	1	1	1	1
11	1	1	1	0
10	0	0	0	0

$$N_2 = (C_2 + C_1 + C_0) \cdot (C_3 + C_0') \cdot (C_3' + C_1' + C_0)$$

3) N_1 :

$C_1 C_0$	00	01	11	10
$C_3 C_2$	00	0	1	1
01	0	0	0	1
11	0	1	1	1
10	0	0	0	1

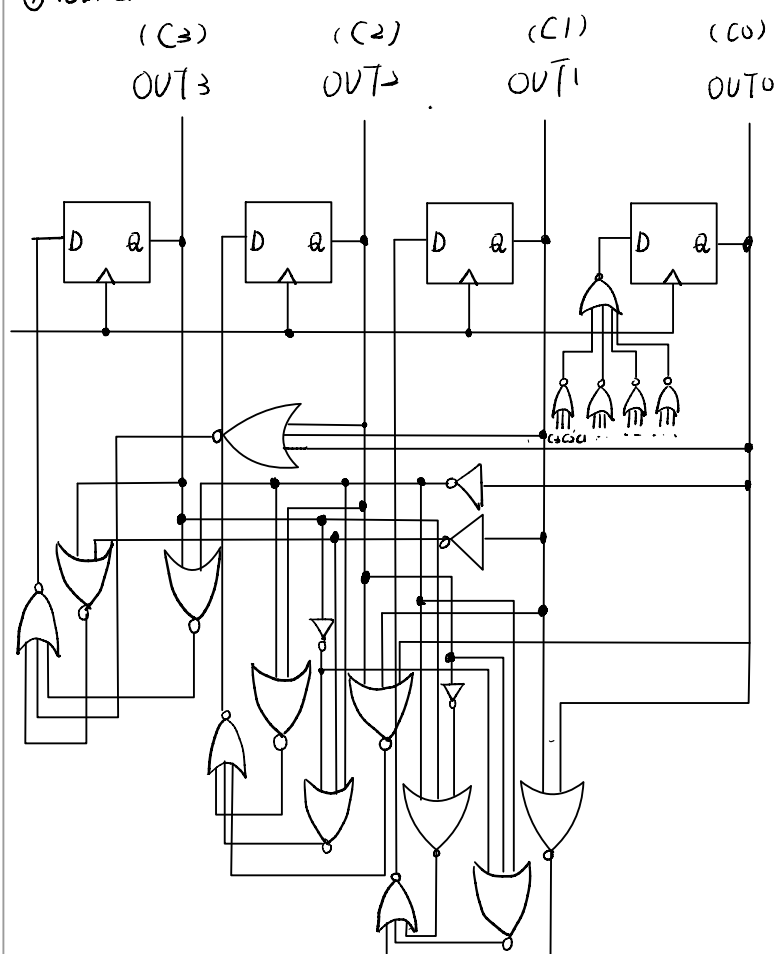
$$N_1 = (C_1 + C_0) \cdot (C_3 + C_2' + C_0') \cdot (C_3' + C_2 + C_0')$$

4) N_0 :

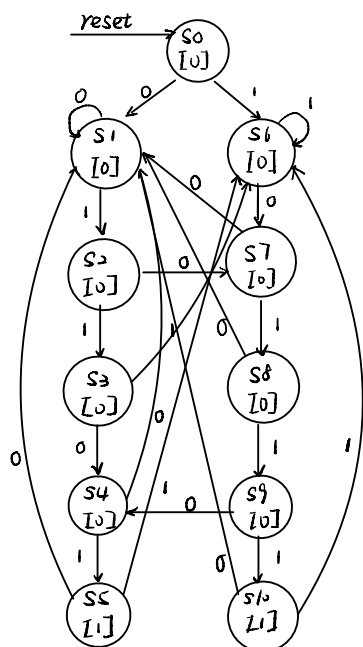
$C_1 C_0$	00	01	11	10
$C_3 C_2$	00	1	1	0
01	0	0	1	1
11	1	1	0	0
10	0	0	1	1

$$N_0 = (C_3 + C_2' + C_1) \cdot (C_3' + C_2 + C_1) \cdot (C_3 + C_2 + C_1') \cdot (C_3' + C_2' + C_1')$$

④ 电路图



2. 解: FSM如下图所示.



3. 解:

